CURRICULUM VITAE

DIPIKA DIGDARSHINI PRADHAN
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Post Applied for: **Post-Doctoral Fellow**

Educational Qualification:

Ph.D (Electronics)	:-	**July 2014-April 2021**
Topic of research	:-	Experimental Analysis of Erbium doped Fiber Amplifier and Design optimization of EDFA, Raman and Hybrid amplifier for DWDM system
University	:-	Sardar Vallabhbhai National Insititute of Techology (SVNIT),Surat
Research Supervisor	:-	Dr. Abhilash Mandloi, Asst. Prof. Electronics Department

M.Tech (Electronics and Telecommunication systems)	:-	**July 2009 – June 2011 (8.72 CGPA)** Driems Autonomous College, BPUT, India First Class

B.Tech (Electronics and communication):-		**July 2001- June 2005**
University	:-	J.I.T.M, Berhampur University, India
Class		First Class

Skill: Optiwave, COMSOL Multiphysics , MATLAB and Python Programming,

List of conferences, journals etc. publications, Workshop:

Publication:	**International Journal:** 1. **D. Pradhan**, Abhilash Mandloi, G.Palai "Efficient hetero amplifier for DWDM system, IJLEO, Elsevier, vol 179,2019,315-322,2019. **[SCI Indexed]** 2. **D. Pradhan**, Abhilash Mandloi ,"Design Optimization of EDFA for16x10 Gbps EDFA DWDM system using different pumping configurations", Wireless personal communication, Springer, vol 102,issue-3,oct-2018**.** **[SCI Indexed]** 3. **D.Pradhan,** AbhilashMandloi," Performance Analysis of flat gain Wideband Raman Amplifier for S+C and C+L band DWDM System",Advances in optoelectronics, **WOS, SCOPUS**, 2018, Article ID

	5703805, doi.org/10.1155/2018/5703805. 4. **D.Pradhan,** AbhilashMandloi," Performance analysis of backward multipumped Raman amplifier in DWDM system", **SCOPUS, Elsevier**, vol 115, page no 182-187,2017 **5. D.Pradhan,** A simplified model and performance optimization of Raman-EDFA hybrid amplifier for DWDM system", Optical and Quantum Electronics, Vol 54,382,2022. **[SCI Indexed]** 6. **D. Pradhan,** Analysis and Review of EDFA,"IJCSN International Journal of Computer Science and Network", Volume 4, Issue 6, December 2015. **International Conference:** 7. D.Pradhan, Abhilash Mandloi," Design Optimization of EDFA in 16x40 Gbps data rate WDM system, **SCOPUS**,doi.org/10.1364,Photonics 2016,Tu4A54.IIT,Kanpur. **8.** D.Pradhan, Abhilash Mandloi ,"Efficient utilization of Er+3-Yb+3/Raman Hybrid amplifier for 40Gbps data rate WDM system, International Conference on Fiber Optics and Photonics 2018,FP003,December12-15,**[SCOPUS Indexed]** 9. D.Pradhan, AbhilashMandloi," Performance analysis of backward multipumped Raman amplifier in DWDM system",**SCOPUS, Elsevier**, vol 115, page no 182-187,2017 10. D.Pradhan, Abhilash Mandloi," Performance analysis of hybrid amplifier in DWDM system at different transmission distance and dispersion",**SCOPUS, Springer**, LNICST, volume 220,Jan,2018 **11.** D.Pradhan,Abhilash Mandloi,"Parabolic Pulse Generation at 1550nm Raman amplifier Utilizing High Power Pump Laser", **SCOPUS, Springer** 12. Pradhan,Sabita Mali, Design and Implementation of Text and Audio Signal Transmission using Visible Light Communication, International Conference on I-SMAC,**IEEE** (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC).
Patents	**1. D. Pradhan,** An Efficient Gain Flattened Erbium Doped Fiber Amplifier for DWDM system, **application no. 202021008827.** **2. D. Pradhan,** Hybrid illumination system using fiber optics**, application no. 20191000656.** **3. D. Pradhan**, I Dongle used as pen drive, application no. **20191000781.**
Paper Presented	1. Design Optimization of EDFA in 16x40 Gbps data rate WDM system,**SCOPUS**,IIT,Kanpur. 2. Performance analysis of backward multipumped Raman amplifier in DWDM system", **SCOPUS, Elsevier,** RSET,Kerla. 3. Performance analysis of hybrid amplifier in DWDM system at different transmission distance and dispersion", **SCOPUS, Springer**, SVNIT,Surat. **4.** Parabolic Pulse Generation at 1550nm Raman amplifier Utilizing High Power Pump Laser, **SCOPUS, Springer**,NIT,Jaipur, OWT 2018,LNEE,volume 546,12 April 2019.**[SCOPUS Indexed]**

	5. D.Pradhan, Abhilash Mandloi ,"Efficient utilization of Er+3-Yb+3/Raman Hybrid amplifier for 40Gbps data rate WDM system, International Conference on Fiber Optics and Photonics 2018,FP003,December12- 15,IIT,Delhi. 6. D.Pradhan,"Design of test and audio signal transmission using visiblelight communication,4th international conference on I-SMAC.**IEEE** preceding,303-306,
Student Paper	1. Akshay Dahatonde, Dipika Pradhan," DESIGN OF SINGLE BAND MEANDER MONOPOLE ANTENNA FOR LTE OPERATION, **IETE** MTS 2018, presented. 2. Ashwini Ubale, Dipika Pradhan, Advances in Power, SMART SHOPPING TROLLEY USING RASPBERRY PI, Signal and Information Technology (APSIT) 2021, **IEEE** proceeding, 2021 International Conference in Advances in Power, Signal, and Information Technology (APSIT),2021
Workshop ,FDP Attended:	▪ Attended hands on workshop on Python Programming, organized in association with TTRIX software technologies during Dec 21 2021, to 12th Jan 2022. ▪ Attended FDP on Arduino included a demonstration cum hands on exposure, organized in association with Spoken Tutorial Project, IIT Bombay. ▪ Attended Awareness Program on Digital Teaching Learning: Need of Time organized by Department of Electronics and Telecommunication Engineering, JSPM Narhe Technical Campus during May 25, 2020, to May 30, 2020 ▪ Attended Faculty Awareness Program on"NAAC Revised Accreditation Framework 2020"Organized by JSPM Narhe Technical Campus From 11/05/2020 to 15/05/2020. ▪ Attended Webinar on "Optical Fiber Communication –Career Opportunity in the age Covid-19" held on 13th May, 2020 jointly organised by IEEE Student Branch of JSPM's Jayawantrao Sawant College of Engineering (STB-12401) & JSPM's JSCOE Electronics & Telecommunication Department, Pune-28. ▪ Attended 1 Week HANDS-ON Faculty Development Program on Artificial Intelligence using Python organised by Department of Electronics & Telecommunication Engineering, A.I.S.S.M.S. Institute of Information Technology, Pune Supported by I.E.T.E. Pune Section. ▪ One Week Online Faculty Development Program on "Linux" at IIT Bombay organized under Spoken Tutorial Project, IIT Bombay, funded by NMEICT, MHRD, Govt. of India from 25th May 2020 to 29th May 2020 ▪ Attended Workshops on advances in image and speech processing at JSPM, Tathawade, PUNE ▪ Attended Workshops on advances in image and speech processing at JSPM, PUNE. ▪ Attended Workshops on OPTOELECTRONICS at SVNIT, SURAT ▪ Attended STTP on Advancement in Antenna measurements and metamaterials at SVNIT, SURAT. ▪ Attended STTP on Advancement in optical engineering at SVNIT, SURAT. ▪ Attended Faculty orientation Programme on Teaching methodology (WN) at Bharti Women college of engg by SPPU, PUNE.

	▪ Attended ISTE workshop on Analog Electronics conducted by IIT Kharagpur under the NMEICT (Govt India) MHRD. ▪ Attended Faculty Development Programme on Optical Communication Networks at Indian Institute of Technology, Kharagpur.
	▪ Attended Faculty orientation Programme on Teaching methodology at KJ Colege of Engg by SPPU, PUNE. ▪ Attended Faculty orientation Programme on Teaching methodology at JSPM, PUNE ▪ Attended Workshop on Engineering Applications Using Soft Computing Techniques at Bhubaneswar Engg College, Bhubaneswar. ▪ Attended Workshop on Advanced Signal Processing and Communication (WASPC-2010) at Silicon Institute of Technology. ▪ Attended International Symposium on Computing Control & Instrumental Application (ISCCIA-2011) at Bhubaneswar Engineering College, Bhubaneswar. ▪ Attended Participated two-day Workshop on "Fundamentals of Electronics & Telecommunication Engineering" under ISTE Local Chapter. ▪ Participated two-day Workshop on "Advanced Digital Signal Processing" organized by JSPM's RSCOE, Department of E&TC Engineering. ▪ Participated two-day Workshop on "Advanced Digital Signal Processing & its Applications" organized by JSPM's BSITR(W), Department of E&TC Engineering. ▪ Attended "Workshop on Teaching Methodologies for Basic Electronics Engineering-FE 2012" organized by SITS Narhe,
Short Term Training Program	ISTE workshop on Analog Electronics conducted by IIT Kharagpur under the NMEICT (Govt India) MHRD.
Reviewer	International Journal of Electrical and Computer Engineering ECE, **SCOPUS**,2017 International Journal of Engineering Research and Reports, peer reviewed journal,2020
Invited Talk	• As a resource person topic in informed search strategy and prolog structure on 8/01/2022 at **NCER**, Pune, Computer Engineering Department-Artificial Intelligence. • As a resource person in TEQIP- III, BPUT Odisha on 15/01/2021 at **GITA** Bhubaneswar. • As a resource person for Antenna and wave propagation in **BSCOER**, Pune.

Workshop/Seminar Organized:

- Coordinator for organizing webinar on," Webinar on Significance of DAQ and communication techniques in ocean energy",NIOT Chennai.
- Coordinator for organizing guest lecture on "being winner with power of thought",JSPM ,NTC.
- Convener for organizing seminar on IOT under IETE,JSPM,NTC
- Inter college Mini Project competition as coordinator under Techno vision,SPPU,Pune University.
- Add on courses on mat lab and Simulink as Coordinator from the Department,JSPM,NTC.
- Add on courses on Animatronics as Coordinator from the Department,JSPM,NTC
- Add on courses on Embedded System Design as Coordinator from the Department,JSPM,NTC

Courses:

- NPTEL certification courses.
- 1-Fiber optics communication
- 2-Product design and Development.
- 3- Outcome Based Pedagogic Principles for Effective Teaching

Online Courses:

- Scilab online courses certification,IIT Bombay
- MATLAB and Simulink in Engineering Education,IIT Bombay
- Experimentation and Modeling,JSPM,NTC

Teaching Expertise:

- Artificial Intelligence
- Machine Learning
- Electromagnetics
- Power Electronics
- Broadband communication system
- Antenna and wave propagation
- Wireless Networks
- Digital circuits
- Basic Electronics

Research experience:

- Chemical synthesis of Nanomaterials, Centre of excellence in Nanoscience and Technology for development of Sensors,Berhampur University,
- Green synthesis of Nanomaterials
- Design optimization of Erbium doped fiber amplifier (EDFA) and Hybrid amplifier,DIAT Pune
- Synthesis of silver nano particles, triangles, rods, and discs
- Green Synthesis of silver nanoparticle for glucose sensing

Project Details:

- Laser based communication link,BPUT,Odisha
- Classification of ECG signals by using Fuzzy c means and k means algorithm,BPUT,Odisha

Project carried out for BE students:

- IOT based health monitoring system
- IOT based water quality monitoring system
- Smart iot car for accident prevention
- Attendance monitoring system using image processing
- Image compression using DCT.
- Smart antenna.
- Design of 16 bit ALU and its implementation in FPGA.
- Android based patient monitoring system.
- Solar tracking system.
- Smart traffic Assistant.
- Smart trolley shopping using Raspberry pi.
- Mini projects-14.

Portfolio:

- IETE students' Forum staff in-charge for the Department.
- Active participation as NAAC criteria 6 and IQAC Coordinator for the Department.
- Active participation as a Add on courses, Enrichment courses and skill development courses Coordinator for the Department
- Active participation as a Project Coordinator for the Department.

Involvement in certification:

- Worked for the ISO 9001: 2008 Certification for the College
- Worked for the NBA Certification for the Department.

Research Interest:

Fiber Optics
Nanomaterials
Photonic biosensor
Wearable sensors
Photonic waveguides

Management and Leadership Skills:

Analytical Decision Making
Teamwork
Adaptability
Trustworthiness

Hobbies and Interests:

Research Playing
Mentoring and Coaching Reading
Watching Television

Language skills mother tongue(s): Odia

Other language(s): English, Hindi

Membership:

- Lifetime member of Indian Societyof Technical Education (MISTE).
- Lifetime Associate member of The Institutions of Electronics and Telecommunication Engineers (IETE).
- Lifetime member of Optical Society of America (OSA)

References:

1. Dr. Abhilash Mandloi
   Asst. Professor
   Sardar Vallabhbhai National Institute of Technology, Surat
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   Ph.no:7016786449
2. Dr. M.M.Sardeshmukh
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3. Dr. G. Palai
   Professor
   Electronics and communication Engineering & Dean R & D,
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   Head of the department.
   Ph.No:9439045946
4. Dr. Vijay Janyani
   Professor
   Dept. of Electronics & Comm. Engineering, Malaviya National Institute of Technology
   MNIT, Jaipur, India
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